Hostel Management System

Deviations from SRS, SAD and Test Plan

11 November 2026. Team: Krish Mathur, Atharva Mihir Gore, Samyuktha, Tanishk Gowda.

1. Purpose

Deliverables Part-2 item 2.i asks for deviations from the submitted SRS, SAD and Test Plan to be documented and explained at the final demo. This document lists every known deviation, what was planned, what was built or done instead, and the effect.

2. Deviations

#	Area	Deviation	Planned	Actual	Impact
1	Process	Development ran into a third sprint	Plan: two sprints (19-23 Oct, 26-30 Oct).	Room allocation, modify, cancel, history (HMS-F-010 to F-014, F-018) and check-in / check-out (F-015, F-016) were finished in Sprint 3 (2-6 Nov), which Deliverables Part-3 allows for spill-over.	None: all 19 functional requirements are built and tested.
2	Process	One combined demo video	Part-2: one video for Sprint 1 and one for Sprint 2.	A single video covers both sprints, captioned by sprint and requirement ID.	Same content in one file.
3	Design	Student landing page	SAD: students land on their own home page after login.	Students land on Room availability; their profile is one click away in the navbar.	Minor navigation change.
4	Design	Hostel fields	STP UT_HOS_01 data: Type and Warden.	Hostel has name, code, total capacity, address and notes; warden can be recorded in notes.	UT_HOS_01 data revised.
5	Design	Availability API response	SAD 3.3.3 example shows 6 fields.	Implemented as designed (room_id, hostel, room_number, capacity, active_allocations, available). The STP case IT_AVL_01 lists only 3 of them; the case was updated.	None.
6	Design	Logging to a file	SAD 3.4: logs/hms.log with INFO, WARNING and ERROR.	Services call Python logging with IDs only, but no file handler is configured, so events go to the console.	ST_SEC_01 log search not applicable (D-04).
7	Design	Deployment hardening	STP 5.1: check --deploy before the demo; HTTPS cookies.	Secure cookies are on when DEBUG=False. HSTS and SSL redirect are not set because the demo is not served over HTTPS.	2 deploy-check warnings (D-05).
8	Testing	System tests automated with the Django test client	STP 5: manual browser runs, then Selenium.	ST cases drive the real URLs, views, templates and database through the Django test client and run in CI.	Repeatable on every commit; no browser rendering check (covered by the demo).
9	Testing	Performance test without Locust	ST_PERF_01: Locust, 20 users, 5 minutes.	In-process load test: 160 requests over 4 pages on 500 students and 100 rooms; p90 31 ms.	No concurrent-user load; result far inside the 3 s target.
10	Testing	Restart test as two processes	ST_REL_01: stop and restart the server.	One process writes records to a file database and exits; a new process re-opens it and verifies them.	Same check without a web server.
11	Testing	Usability walk-through not yet run	ST_UX_01: 3 students who did not build HMS.	Scheduled for the UAT demo.	HMS-NF-004 partly validated until then.
12	Testing	Dashboard compared with ORM counts	ST_DASH_01: compare with DB Browser for SQLite.	Automated test compares each displayed value with a direct database count.	None.

#	Area	Deviation	Planned	Actual	Impact
13	Testing	Test cases added	STP: 58 cases.	7 cases added for behaviour built in the sprints (check-in POST only, check-out confirmation, repeated check-out, end-to-end fuzzing, error pages, CI pipeline). Total 65.	Better coverage.
14	Scope	Student home allocation card	SAD 3.5: allocation card on the student home page.	Added after Sprint 3 together with a layout tidy-up.	Now as designed.

3. No change to scope

All 19 functional requirements, 6 non-functional requirements and 6 security requirements in the SRS are implemented. Nothing outside the SRS scope (fee payment, attendance, visitor management, integration with other college systems) was added.